

ICT702 Data Visualisation

Introducing Data Visualisation Module 1





I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



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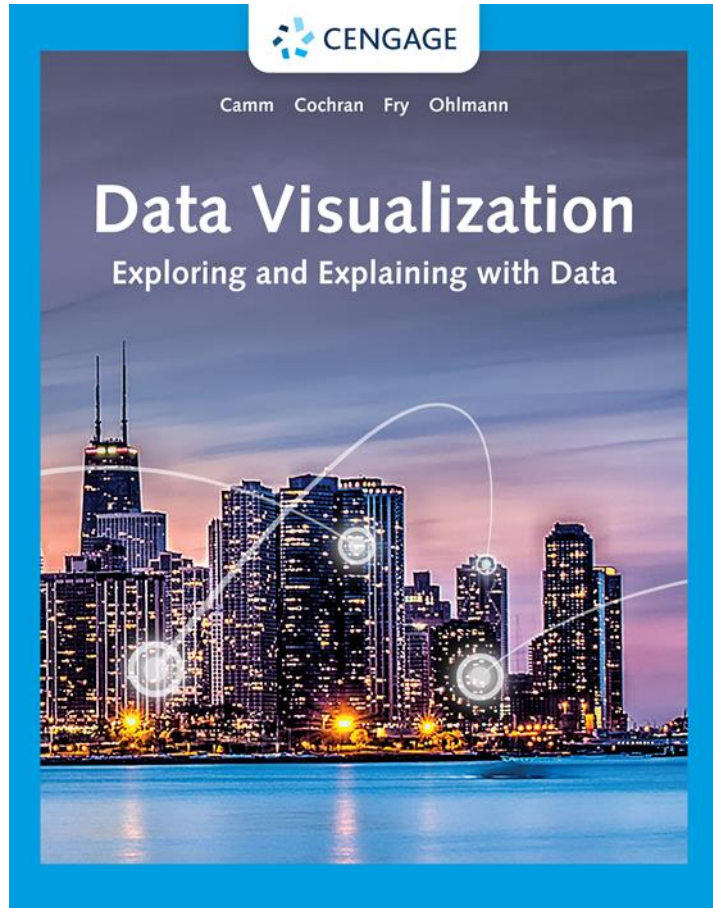
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Course text



Camm, Cochran, Fry & Ohlmann,
Data Visualization

- Exploring and Explaining with Data,
1st Edition. Cengage

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Data Visualisation

The graphical representation of data and information using displays such as charts, graphs, and maps.

Learning Objectives for Module 1

- Define analytics
- Describe different types of analytics
- Use examples to describe types of data
- Describe data visualisations used in practice
- Identify various types of charts

Part 1

Analytics

Analytics is the scientific process of transforming data into insights for making better decisions.

Three developments have spurred the explosive growth in the use of analytics for decision making:

- **Technological**
- **Methodological developments**
- Explosion in **computing power** and **storage capability**

A Categorisation of Analytical Methods

- **Descriptive analytics** – encompasses a set of analytical tools that describes what has happened in the past.
 - Examples: data queries, standard reports, descriptive statistics, data visualisation, cluster analysis, and what-if spreadsheet models.
- **Predictive analytics** – consists of analytical tools that use models constructed from past data to predict the future or ascertain the impact of one variable on another.
 - Examples: linear regression, time series analysis, and predictive data mining.
- **Prescriptive Analytics** – includes mathematical or logical models that indicate the best course of action to take in decision making.
 - Examples: optimization, simulation, and decision analysis.

Why Visualise Data?

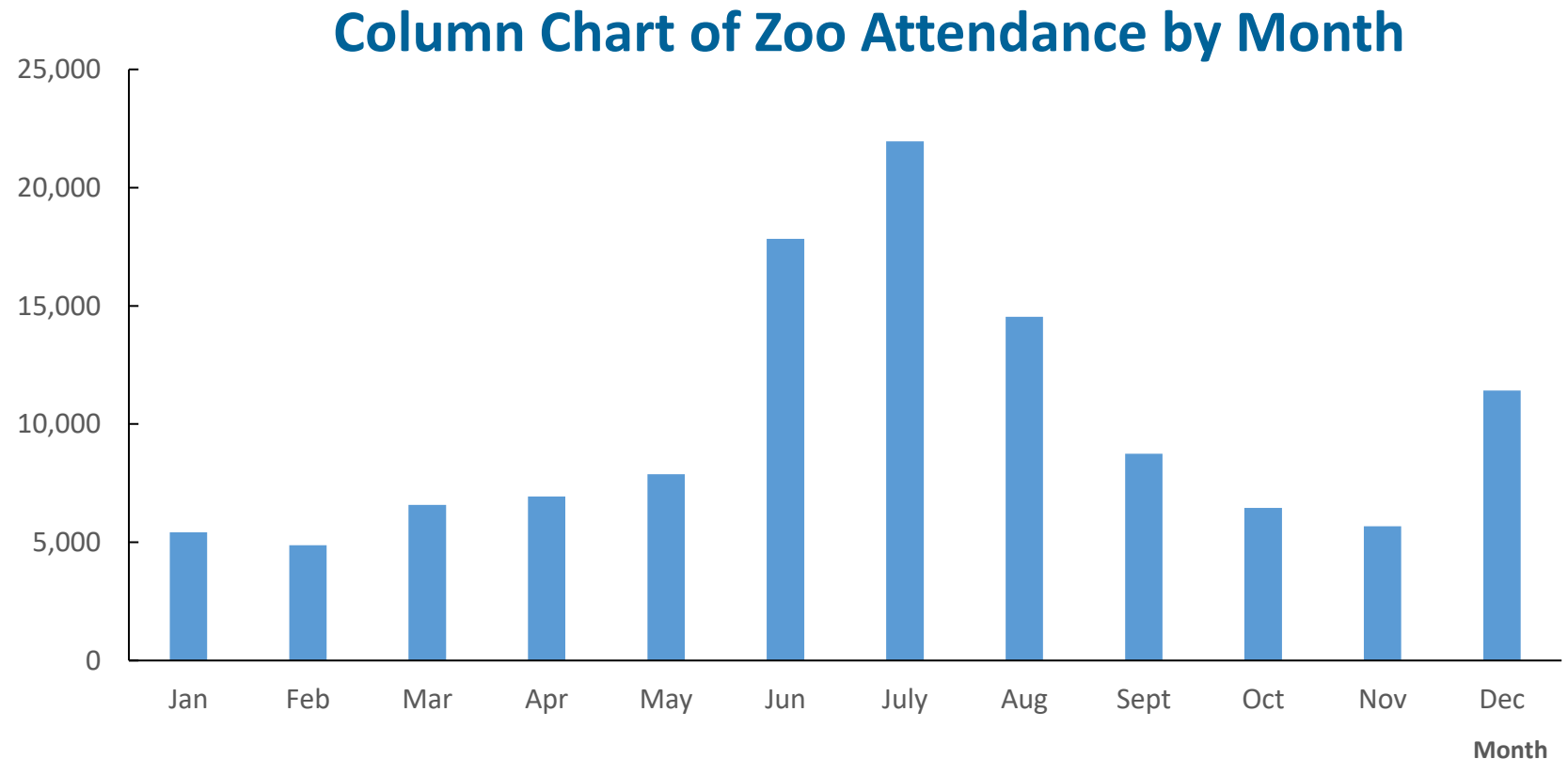
We create data visualisations for two reasons:

1. Exploring data
2. Communicating/explaining a message

Data Visualisation for Exploration: Patterns

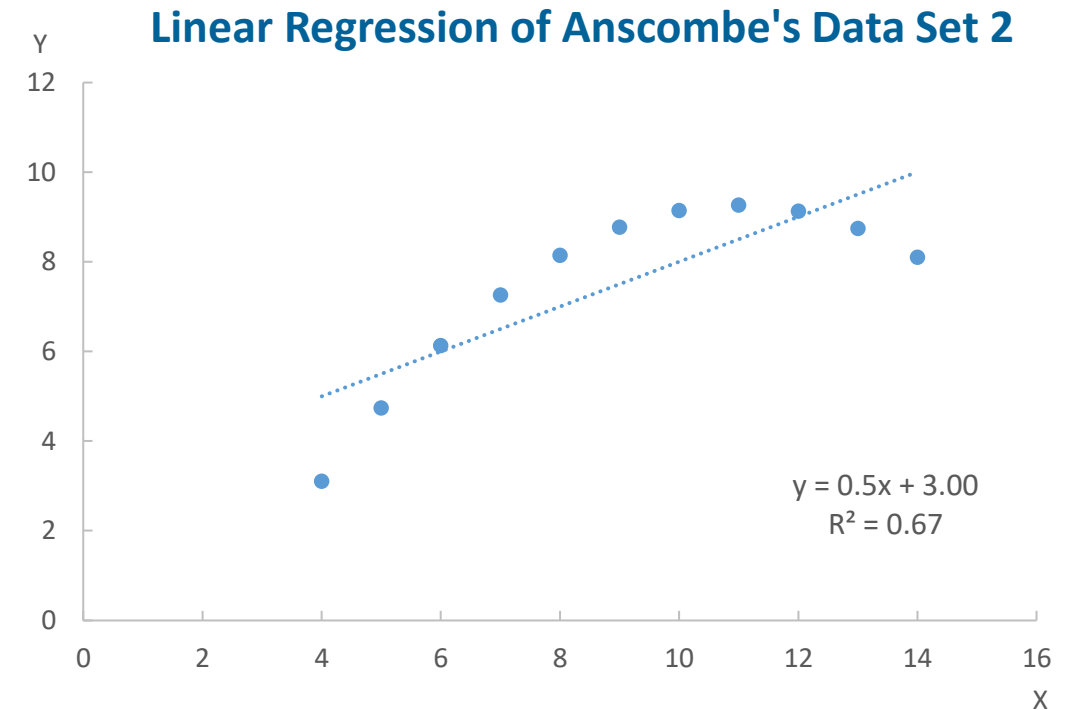
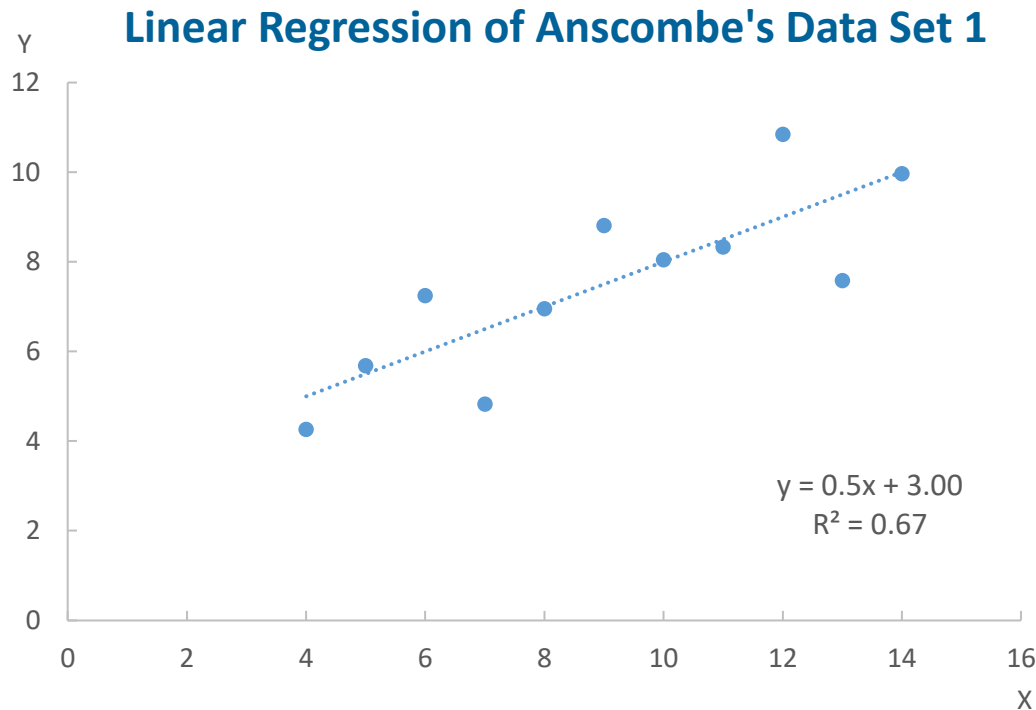
Data visualisation is a powerful tool to identify patterns.

Month	Zoo Attendance
Jan	5,422
Feb	4,878
Mar	6,586
Apr	6,943
May	7,876
Jun	17,843
July	21,967
Aug	14,542
Sept	8,751
Oct	6,454
Nov	5,677
Dec	11,422



Data Visualisation for Exploration: Relationships

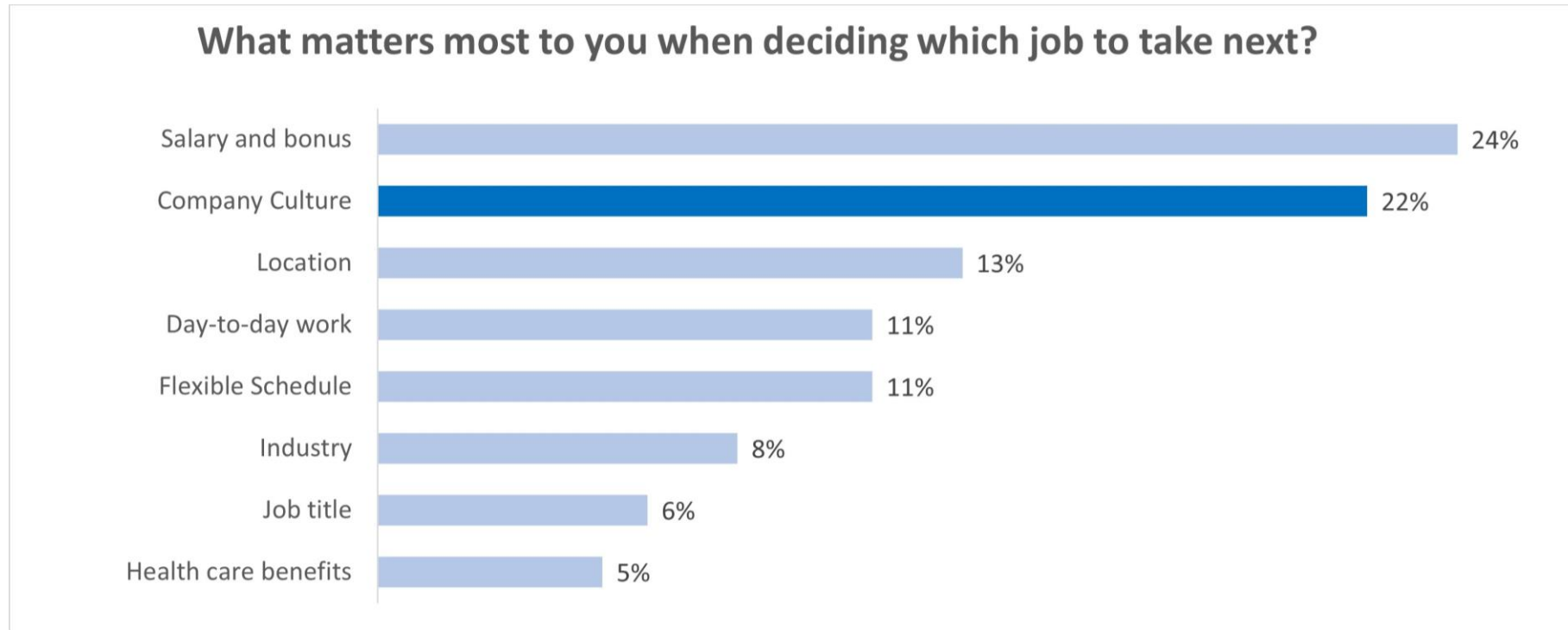
Data visualisation is a powerful tool to better understand the relationship between variables.



Data Visualisation for Explanation

Data visualisation is important for explaining:

- relationships found in data
- results of predictive and prescriptive models.



Quantitative vs Categorical Data

Data for the Dow Jones Industrial Index Companies (April 3, 2020)

Company	Symbol	Industry	Share Price (\$)	Volume
Apple Inc.	AAPL	Technology	241.41	32,470,017
American Express	AXP	Financial Services	73.6	9,902,194
Boeing	BA	Manufacturing	124.52	36,489,379
Caterpillar Inc.	CAT	Manufacturing	114.67	4,803,174
Cisco Systems	CSCO	Technology	39.06	21,235,157
Chevron	CVX	Petroleum	75.11	14,317,998
Disney	DIS	Entertainment	93.88	14,592,062
Goldman Sachs	GS	Financial Services	146.93	2,773,298
Home Depot, Inc.	HD	Retailing	178.7	6,762,357
IBM	IBM	Technology	106.34	3,909,196
Intel Corporation	INTC	Technology	54.13	23,906,062
Johnson & Johnson	JNJ	Pharmaceutical	134.17	9,409,033
JPMorgan Chase	JPM	Financial Services	84.05	20,363,095
Coca-Cola	KO	Food	43.83	13,294,556
McDonald's	MCD	Food	160.33	4,361,094
3M Company	MMM	Conglomerate	133.79	3,461,642
Merck & Co.	MRK	Pharmaceutical	76.25	9,181,539
Microsoft	MSFT	Technology	153.83	41,243,284

Quantitative data

Data for which numerical values indicate magnitude. Arithmetic operations, such as addition, subtraction, multiplication, and division, can be performed on quantitative data.

Examples: Share Price (\$), and Volume.

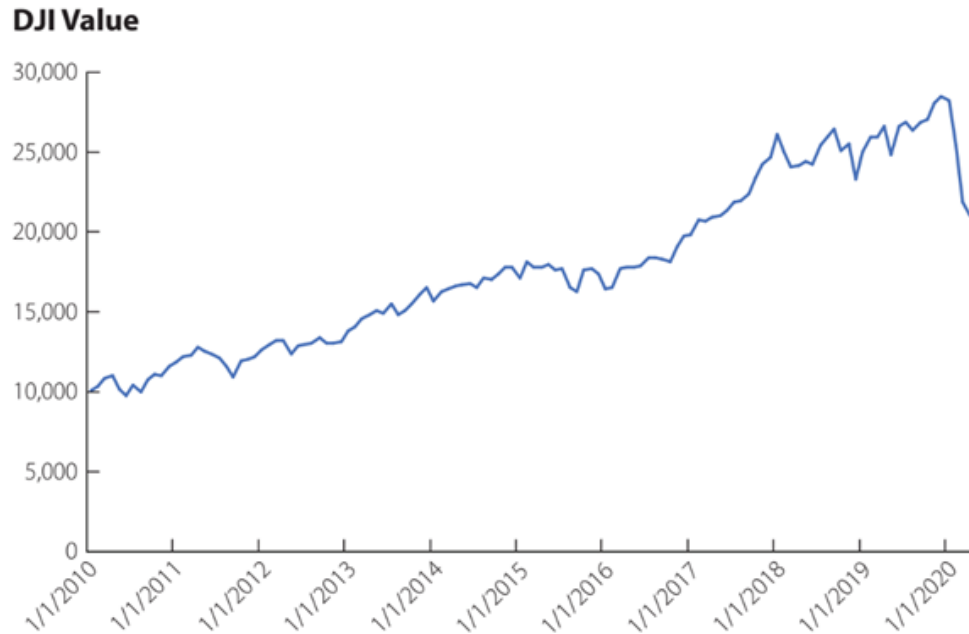
Categorical data

Data for which labels or names identify categories of like items. Arithmetic operations cannot be performed on categorical data.

Examples: Company, Symbol, and Industry.

Cross-Sectional vs Time Series Data

Dow Jones Index values from January 2010 to April 2020



Cross-sectional data

Data collected from several entities over the same time frame.

Time series data

Collected over several time periods.

- Business and economic publications frequently include graphs of time series data.
- Graphs help analysts understand what happened in the past, identify trends over time, and project future levels for the time series.

Big Data

- There is no universally accepted definition of big data.
- A working definition of big data:
 - **Any set of data too large or too complex to be handled by a desktop computer.**
- The four Vs of big data:
 - **Volume** - the amount of data generated
 - **Velocity** - the speed at which the data are generated
 - **Variety** - the diversity in types and structures of data generated
 - **Veracity** - the reliability of the data generated

Word Cloud



A word cloud is a frequently used chart to summarize words used in large sets of text data.

- Source: <https://milady.cengage.com/blog/using-word-clouds-classroom>

End of Part 1